Development of Comprehensive, Data-Driven Workflows to Accelerate and Refine Therapeutic Development

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[1]

Abstract—The pharmaceutical industry currently faces a critical issue in drug development, known as Eroom's Law, indicating a 90% failure rate and a massive slowdown in new drug development despite technological growth. Current toxicity prediction often relies on "black-box" AI or traditional QSAR methods that are not explainable. Hence, this research proposes a quantitative experimental framework using Explainable AI (XAI) models, Random Forest and XGBoost, integrated with SHAP. By comparing these against baseline QSAR models through Tox21 and ClinTox datasets, the study aims to compare their predictive accuracy and explainability. The anticipated impact is that future researchers can use a more refined and accountable drug discovery pipeline that reduces preclinical costs by predicting toxicity of drug candidates in early developmental phases to avoid time and capital waste on unviable candidates.

I. Introduction

A. Background and Rationale

Background and Rationale In the contemporary therapeutic development context, the Eroom's law (opposite of Moore's law). Its argument is that despite technological advancements the pace and cost of developing new drugs double approximately every nine years, based on pharmaceutical industry data since the 1960s [1]. There are multiple factors that drive this law, with most pertinent to workflows being the regulatory pressure and the issue of new drugs only offering incremental improvements over successful and cheap treatments that already exist.

B. Practical, Scientific and Societal Cost

Predictive models are frequently restricted to serious limitations when they are used with real-world bioscience data, even though they are increasingly adopted [3]. Most of the models are very precise at the point of training and testing but do not continue the same performance when presented with new and unknown data. This problem is usually related to overfitting and lack of generalisation, which decreases the validity of such models in real world. This means that the models that seem to be effective in controlled settings might not be effective in actual practice in the clinical setting [4].

[1]GitHub link:

II. CRITICAL LITERATURE REVIEW

A. AI-driven Workflows

Artificial Intelligence (AI) has become a novel emerging technology with deep applications in therapeutic development. According to Zhavoronkov, Gennert, and Shi [3], AI technologies enable a 'prompt-to-drug' pipeline, where AI prompts can help provide novel hypotheses and designs as well as synthesis, validation, and clinical planning. This means that generative AI tools like ChatGPT can provide detailed plans to develop a comprehensive therapeutic development system. These AI tools (LLM- Large Language Models) have the ability to take instruction inputs in order to generate biology models as well as therapeutical practical plans, as well as the ability to evaluate the results to provide integrated insights from the results.

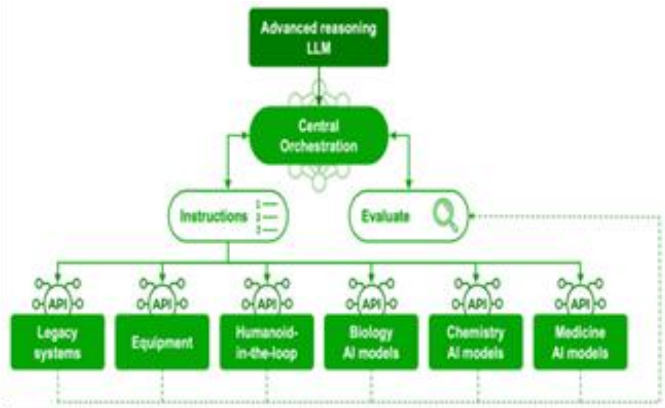

Fig. 1. AI-Driven Workflows [3]

The diagrammatic representation of the AI-driven workflows in this regard is given below. For the evaluation of the articles, the CRAAP test (Currency, Relevance, Authority, Accuracy, and Purpose) has been done. Based on the evaluation, this article provide the latest insights on AI applications in therapeutical development from an authoritative journal, thus providing accurate insights that are relevant to the purpose of this study. The same framework has been used to evaluate the core papers that have guided the development of the main goals of this study as well as the methodology that will be implemented in order to satisfy the objectives developed based on the research gaps that have been found through the literature review.

From the above figure, it can be noted that advanced reasoning AI models can offer optimisation of therapeutic development workflows by constantly monitoring and finetuning operations through the utilisation of real-time data readouts as well as simultaneous evaluation on a real-time basis as well. This illustrates the value of AI tools within the context of Therapeutic development, as optimisation of these workflows help in accelerating and refining the processes. Meanwhile, Mishra, Mishra and Agarwal [4] have argued that implementing only AI in therapeutical development workflows can limit outcomes, and instead focus should be given on combining AI with Internet of Things (IoT), and Robotics, as it enables the formation of an intelligent ecosystem, which is capable of conducting autonomous experimentation, combined with self-correcting and self-improving features, allowing for continuous enhancement of workflows. This is particularly relevant because such an autonomous system can use real-time data acquisition, adaptive control, and predictive modelling across multiple scientific domains, making it valuable for therapeutic development as well as other domains of bioscience. The critical benefit of such a system over only-AI system is that it serves as a self-driving laboratory, which can enhance discovery cycles while minimising experimental uncertainties and errors, thus accelerating therapeutic development as well as helping to refine the workflows.

Moreover, integrated computational and experimental workflows have helped in enhancing reproducibility and optimisation of complex systems as well as enhancing real-time decision making [4]. These aspects highlight the value of digital technologies in enhancing scientific experimentation, which is central to the therapeutic development processes. The overall trend that has been found is that digital technologies have helped to transition from manual to autonomous experimentation, which is the main driving force towards accelerating and refinement of experimentation workflows. In addition to these aspects, there is also the evolution of digital technologies in scientific research, which has led to enhancement of experimentation outcomes. For instance, it can be noted that the most recent advancement in technologies of scientific relevance, there are big data analytics, cloud computing, AI/ML (Machine learning), and IoT sensors, which have helped in accelerating data-driven discovery, as well as automation of experimentation and predictive modelling [4].

B. AI/ML Methodologies in Therapeutic Development

Stenfors [5] has developed a novel system to avoid trialand-error-based experimentation by using computational modelling to predict experimental outcomes by using machine learning models like XGBoost to enable faster prediction of results using existing data. The particular value of such a method is that it is highly scalable, meaning that once set up, it can continuously generate prediction results to automate experimentation without needing manual trial-anderror, and the range of experimentation can be adjusted to increase or decrease it based on the scope of work within a therapeutic development project. This methodology is a novel method to avoid tedious trial-and-error experimentation, thus avoiding 'busy-work', where instead scientists can focus on issues that require human intelligence, thus accelerating and refining the workflows. However, there is the limitation that the training data may not be sufficiently detailed to provide accurate prediction for the machine learning models [5]. Hence, it is critical to thoroughly evaluate the training data to ensure that accuracy is maintained through this approach. The CRAAP evaluation has revealed that this article provides critical application of AI/ML framework for the purposes of chemical reaction prediction, which is of relevance to this study focusing on toxicity prediction. Further, the insights are current and aligned with the purpose of this study. In alignment with this perspective, Ahmadi [6] has found that machine learning has enabled a revolution in drug discovery, allowing scientists to quickly identify prospects of therapeutic value by analysing the intricate chemical structures in order to predict how drugs will interact with their targets as well as predict pharmacological profiles. This is valuable because it helps in accelerating the creation of new therapies and treatments, which is the central goal of therapeutic development. Therefore, application of machine learning is critical for this purpose, as it has become a new standard of experimentation in such projects. In particular, the first area of application is that of target identification and validation, where the precise molecular targets need to be identified. In alignment, Meng, Qin, Wang and Zhao [7] have argued that machine learning algorithms examine comprehensive datasets of biological, genetic, and clinical data that help in identifying undiscovered connections throughout intricate biological networks, which helps researchers be guided to strong possibilities. Similarly, machine learning algorithms also enable determination of whether the target is biologically relevant and likely to respond to therapeutic intervention, thus enabling target validation. The CRAAP evaluation has revealed that both these articles provide the latest developments in AI/ML applications in therapeutic development, justifying their inclusion in the literature review. In addition, machine learning algorithms can also examine such data in order to find possible candidates of drug repurposing or repositioning by evaluating relationships among medications, illnesses, and biological circuits in order to determine ailments that may be treated with a drug under scrutiny. In alignment Mehran et al. [8] have found that it enables machine learning algorithms to detect existing medications with the potential to treat various ailments by finding patterns within the examined dataset. Hence, effective drug repurposing is achieved through this methodology. On the other hand, Nada et al. [9] have asserted that the impact of machine learning on decreasing therapeutic development costs come from its application in High-Throughput Screening

(HTS). In this regard, machine learning approaches like clustering and classification of algorithms can be used to quickly evaluate compounds based on their structure, characteristics, and biological data in order to determine the ones with sought-after therapeutic effects. It aids in minimising the time and costs associated with manual HTS, thus lowering the overall investment requirement for new drug development. As per the CRAAP evaluation, these papers provide details of the underlying mechanisms by which machine learning algorithms work in therapeutic development context, thus making them relevant, current, and authoritative as well as aligned with the purpose of this study. In addition, Borkotoky, Joshi, Kaushik and Jha [10] have argued that predictive toxicology is another critical aspect in therapeutic development, which involves evaluation of potential toxicity of new drug candidates in the early developmental processes. This can enable minimisation of wastage of capital on seeking development of drugs that may be overly toxic to be practical. Here, deep neural networks and random forest are two prominent algorithms that can effectively capture intricate correlations between chemical characteristics and toxicity, allowing them to act as practical predictive toxicology models. This has major implications because it helps in removing toxic therapeutic candidates before expensive preclinical and clinical trials, thus minimising costs and avoiding time wastage due to the same. On the other hand, Malheiro, Santos, Figueiras, and Mascarenhas-Melo [11] have asserted that machine learning has applications in clinical trial optimisation as it can enable recruitment of suitable participants by evaluating EHR (Electronic Health Records), thus enhancing recruitment phase as well as improving classification of qualified participants to enable quick starting of the clinical phases. Moreover, the study has found that machine learning can help in predicting clinical trial outcomes, which can be compared with real-time results to determine whether to continue or stop the trials, thus saving time and money on failed projects. Therefore, there is a high potential of streamlining the clinical phases, allowing for quick ending of projects that are not likely to generate positive results, thus saving the time and money to be invested in other possible therapeutic developmental projects. As per the CRAAP evaluation, these papers provide critical details of the application of random forest and general machine learning algorithms to enable better prediction outcomes, which is aligned with the purpose of this study and represents accurate, current, and relevant data.

In overall terms, Ahmadi [6] have argued that AI/ML has brought a new era of healthcare, which is expected for further grow with technological advances. Potential future improvements include early illness detection through improved spotting of biomarkers and patterns identification enabling better preventative measures to be taken. Further, the technology has the potential to greatly boost the pace of therapeutic development as well as minimise costs, which is only likely to improve in the future. Meanwhile, Nada et al. [9] have asserted that with the combination of technological advancements in machine learning as well as wearables, it is likely that in the future telemedicine and remote patient monitoring will improve to the point that the burden on healthcare institutions can be minimised, which can improve overall public healthcare outcomes. This is particularly relevant from a policy formulation perspective, as promotion of machine learning in this domain can help improve governmental efficacy with regards to public health, thus having a positive impact in society.

C. Predictive ADMET

According to Ibrahim, Isah, Uba and Uzairu [13], Quantitative Structure-Activity Relationship (QSAR) modeling and toxicity prediction are essential components in ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity). The article focuses on implementing a computational drug design approach by utilising genetic function approximation multilinear regression to achieve high predictive accuracy for inhibitory activity. Further, the combination of QSAR modelling, molecular docking, ADMET analysis, and DFT calculations in order to achieve high model efficacy [13]. The research highlighted the value of QSAR modelling within the context of ADMET prediction of compounds, which is a critical component of therapeutic development as managing these relationships is critical to ensure that new drugs are safe and effective to use. Accordingly, there is the relevance ensuring that safety of drugs is maintained, which ties to the compliance of healthcare regulations as well as effectiveness in terms of generating the intended therapeutical effect in the body. On the other hand, Gupta et al. [14] utilised Molegro Virtual Docker to conduct molecular docking investigations on target proteins involving computational tools such as pkCSM and ADMET profiling for developing potential therapeutic strategies against HIV/AIDS. As a result, potential compounds for HIV mitigation were identified, focusing on the value of structural analysis in understanding ligand-and-receptor interactions for drug design in a rational manner against HIV. This also illustrated the value of computational models for therapeutic development, focusing on the context of developing drugs to tackle HIV/AIDS. In combination of these two articles, it can be noted that computational modelling has considerable significance of applications for enhancing predictive ADMET in order to identify safe and effective new drugs that may be used in future therapeutic treatments. Apart from these aspects, Verma and Kumar [15] have found that AI methods such as deep docking, active learning, and multi-task learning have greatly impacted therapeutic developmental processes by reducing the timeline for virtual screening and hit identification. However, the study also noted that the theoretical basis for such impacts shows much higher promise that what is actually recorded in real-world cases. The article also found that by July 2025, more than 29 publicly-reported AI-based therapeutic programmes are already advanced to

human studies. This illustrates the practical relevance of AI in enhancing therapeutical development programmes. In particular, the study notes that deep learning models like Graph Neural Networks (GNN) and transformers have the ability to generate a much higher accuracy to detect molecular property and ADMET prediction in comparison to traditional QSAR methods. Accordingly, there is an evolution towards AI/ML methods in the present times, and this trend is illustrated in the figure given below. These papers provide critical insights pertaining to traditional QSAR methods for toxicity prediction, making them highly relevant, authoritative, current, and aligned with the purpose of this investigation.

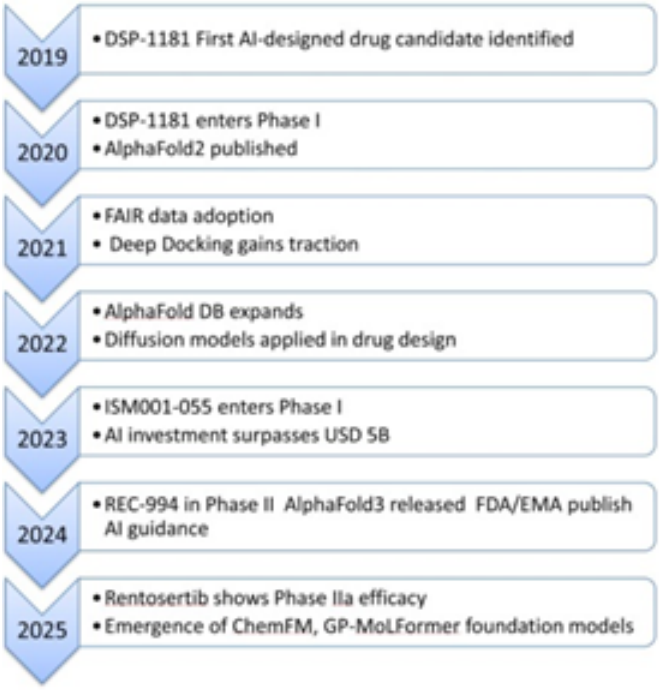

Fig. 2. AI/ML-Enabled Drug Discovery Pipeline

From figure 3, it can be noted that AI/ML-based applications began with DSP-1181, which was the first AI-design drug candidate and this happened in 2019, whereas latest developments as of 2025 involve emergence of ChemFM and GPMoLFormer foundation models for therapeutic development [15]. Meanwhile, other studies [13] [14] have illustrated the relevance of traditional computational QSAR methods in enhancing ADMET predictive accuracy. This means that AIbased ADMET prediction has much higher accuracy, indicating that future efforts in this direction are likely to move towards AI and ML-based ADMET prediction models through frameworks like GNN.

D. High-Throughput Experimentation (HTE)

HTE represents the application of automation features in order to develop self-driving laboratories. In this regard, Nsouli, Galiyan and Laura [16] have explained that HTE involves creation of a workflow where the AI is responsible for generating ideas for therapeutic development, which it then carries out through robotics technology, which in turn supplies the AI algorithms with the resultant data in order to refine the next step or move to another therapeutic candidate. This represents a fully autonomous laboratory that is able to operate without human intervention, thus boosting the overall efficiency of experimentation and can be used to establish active learning loops to enable machine learning that can help in effective iterations to be run until viable therapeutic solutions can be found. The study has also noted the significance of Bayesian optimisation in such frameworks. Meanwhile, the results highlighted that plate design optimisation is critical as it significantly improves data reliability as well as reducing spatial bias [16]. This helps in ensuring that errors are avoided as the autonomous loops will extrapolate these errors, leading to grave inaccuracies in the final results, thus making the entire framework unviable.

Apart from these aspects, Nada et al. [9] have found that there is the need of implementing a flexible yet standardised workflows for diverse chemistries, as there is the need of ensuring autonomy to the system in order to ensure its continuous learning loops are effective. Here, the main considerations involve improvement of reproducibility, precision, and scalability along with closed-loop optimisation with real-time analytics. It is through these experimental execution considerations that HTE is able to generate effective results high a high degree of accuracy and viability for real-world applications. This is in alignment with the expectations emergent from the results of multiple studies [16] [13]. Further, there is the relevance of high-throughput analysis with an HTE framework. As such, technological advancements such as MALDI-MS, DESI-MS, and ADE-OPI-MS, representing ultra-fast mass spectrometry tools. These advances has the impact of accelerating drug discovery by enabling rapid screening, real-time evaluation, and data-driven optimisation of synthesis pathways, which helps in reducing time-to-market and improving candidate selection efficiency.

III. Research Gap and Objectives

A. Research Gap

The evaluation of the present literature pertaining to datadriven workflows aimed towards accelerating and refining the existing therapeutic development systems have revealed that there is a considerable level of information pertaining to emerging technologies like AI and Big Data Analytics that have massive potential to bring improvements. In this regard, a major finding was the relevance of machine learning algorithms such as GNN, which is a deep learning model. However, frameworks that are already in use are mostly dependent on these ‘black box’ type of AI/ML models. Within the context of therapeutic development, there is the essential need of researchers understanding why a potential drug

candidate is toxic instead of only knowing whether it is toxic or not. These black box AI/ML models represent frameworks that can provide results, but their rationale of decision-making cannot be interpreted in a human manner. Hence, the 'why' element is lost in such black box models.

Accordingly, there is the potential of Explainable AI (XAI) in this field, as it can not only provide accurate results but also explain the rationale behind the results shown. However, there is a lack of research focusing on XAI model application within the field of therapeutic development. In this regard, Malheiro, Santos, Figueiras, and Mascarenhas-Melo [11] have argued that there is potential of XAI in resolving the current technological limitations, through simpler models like Random Forest or XGBoost (as valuable starting points) to predict ADMET properties with high accuracy while also gaining insights about the inherent decision-making rationale.

B. Research Question

What is the extent to which XAI (Random Forest and XGBoost) can provide accurate and interpretable toxicity predictions compared to traditional QSAR methods for therapeutic development?

C. Research Objectives

TABLE I
SMART OBJECTIVES

Specific	To compare XAI (Random Forest and XGBoost) models against traditional QSAR method for toxicity prediction
Measurable	To evaluate accuracy and interpretability metrics for comparison
Achievable	To establish whether XAI framework is better for understanding the causes of toxicity in new drug development than QSAR method
Relevant	To improve efficiency and quality in therapeutic development by implementing explainable AI (Random Forest and XGBoost) in toxicity prediction phases
Timebound	To complete the research within 9 months

IV. PROPOSED METHODOLOGY

This study will involve the impelmentation of a quantitative experimental research design in order to compare the predictive accuracy as well as the explainability of XAI (Explanable AI) models against traditional QSAR methods. For the purpose of the experimentation, the Python tool will be used. The detailed methodology is given below.

A. Data Acquisition and Preprocessing

Firstly, dataset selection will be done, which will involve the consideration of standardised toxicity datasets such as Tox21 or ClinTox datasets. These comprise chemical structures and labelled toxicity endpoints concerning therapeutical drug components. Therefore, these datasets can provide the required information for the model development and testing of their accuracy as well as explainability. This will be followed by the conversion of chemical SMILES (Simplified Molecular Input Line Entry System) strings into numerical data using Molecular Fingerprints (such as Moran or Circular Fingerprints) as well as physical-chemical descriptors. Then, data splitting will be done to partition the data into 80% going towards training whereas 20% going towards testing. This is justified because it will enable the conducting of an unbiased evaluation [16]. Hence, through this methodological approach effective data acquisition and preprocessing can be done in order to satisfactorily conduct the rest of the experiment.

B. Model Implementation

In the experiment, there will be the implementation of a total of three models, which are explained below:

- Baseline (Traditional QSAR): This model will involve a Multiple Linear Regression (MLR) or Partial Least Squares (PLS) model to represent traditional quantitative structure-activity relationship standards that are expected in traditional QSAR models. As a result, the baseline model will be established, against which the rest of the two XAI models will be compared to assess the difference in accuracy of toxicity prediction as well as explainability.
- XAI Model 1 (Random Forest): The Random Forest model is an ensemble tree-based model that is used to capture non-linear correlations [17]. This has a major potential to improve toxicity prediction as it is not only able to derive linear correlations like QSAR models, but also non-linear correlations. Moreover, it can be made into explainable AI, which is the main reason for its inclusion in this experiment.
- XAI Model 2 (XGBoost): The XGBoost model consists of a gradient-boosted decision tree framework optimized for high-performance predictive accuracy [18]. This model is well-established for its predictive accuracy, and the literature review has found its effective relevance in the therapeutic developmental processes. Hence, it is justified to include this model for comparison against traditional QSAR models in this experiment along with the Random Forest model. It also represents an explainable AI tool, which also indicates its relevance to this experimental research.

C. Evaluation Metrics

For the evaluation of the models, evaluation metrics will be used, which are explained in the following bullet points:

- Predictive Performance: The evaluation matrix will be used to analyse the predictive performance of the models involved in this experiment. This includes evaluation of the Accuracy, Precision, Recall, and the Area Under the ROC Curve (AUC-ROC) [19]. These provide effective metrics for evaluating the overall predictive accuracy of

algorithms and models, thus justifying its inclusion in the proposed experimental research.

- Interpretability (The XAI Element): The second area of evaluation pertains to the analysis of the explainability of the models. In this regard, there will be the implementation of SHAP (SHapley Additive exPlanations) or Feature Importance maps. These frameworks allow moving beyond the traditional 'black-box' environment in order to ascertain the rationale for decisions or results given by AI models [20]. In this experimental research, this can enable determination of the specific chemical substructures (such as a specific functional group within chemical compounds) that contribute the most to a 'toxicity' prediction. This can enable effective derivation of the level of explainability of the models, allowing comparisons to be derived in order to satisfy the objectives of the proposed experimental research.

D. Comparative Analysis

The final phase of the experimental research will involve the comparative analysis to be carried out. Here, the main focus will be on comparing the predictive accuracy and explainability of the traditional QSAR method against the XAI models, Random Forest and XGBoost. Hence, the goal of this study is to determine whether XGBoost and Random Forest perform better than traditional QSAR model in terms of their predictive accuracy for toxicity of potential drug therapies. Moreover, comparisons of their explainability will also be done in order to assess whether their results are explainable by humans, regarding the reason why a compound is flagged as toxic, satisfying the research gap

V. Data, Ethics, and Practical Considerations

A. Data Readiness and Quality

In this experimental research, there will be the acquisition of data from Tox21 and ClinTox datasets, which are goldstandard, open-access repositories for chemical toxicity. Therefore, it is clear that the data readiness is high for this study, as these sets are provided in standardized SMILES formats. However, there is the need of a rigorous cleaning phase in order to handle missing values and manage the 'unbalanced data', such as where non-toxic compounds significantly outnumber toxic ones. Moreover, to ensure legal compliance, all data will be sourced from public domains under Creative Commons or similar open-research licenses. This is essential because it helps to ensure that no intellectual property infringements occur during the drug-discovery simulation processes as part of this study.

B. Ethical Red-Teaming

This study will involve the implementation of a proactive 'red-teaming', where the analysis helps to identify two major risks named representation bias and false negatives. It is expected that if the training data lacks specific chemical classes, the AI may develop a blind spot, which may result in it incorrectly labelling a toxic novel compound as safe. From a social perspective, this means that while the data is molecular and does not contain personal identifiers, the 'black box' nature of secondary AI layers within the model could possibly lead to a lack of accountability in decision-making if the model's rationale is not explicitly clear as these biases could greatly hamper accuracy of the results achieved.

C. Technical Mitigations

In relation to the risks that have been identified, the following technical aspects are considered:

- SMOTE (Synthetic Minority Over-sampling Technique): This will be used to address data imbalance and ensure that the model learns the characteristics of toxic compounds as effectively as safe ones.
- Validation through SHAP: This technical consideration will involve every safe prediction for a high-risk chemical class being cross-referenced with a SHAP feature importance to ensure that the AI model is not ignoring known toxicophores (toxic structural fragments).
- Human-in-the-loop (HITL): The final technical consideration in the experimental research will consist of a final verification stage where the XAI's 'rationale' is checked against established biological literature in order to ensure that the AI's logic aligns with real-world chemistry and biology.